Real-Time Pothole Detection With Edge Intelligence and Digital Twin in Internet of Vehicles

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***Abstract*—In intelligent transportation systems (ITSs), computer vision and digital twin (DT) technologies are crucial for enhancing safety and efficiency. High-speed vehicles require timely alert systems to prevent collisions with other vehicles and infrastructure, as even a single misjudgment can lead to severe road accident. Advanced driver-assistance systems (ADASs) and vehicular ad-hoc networks (VANETs) enable vehicle-to-vehicle cooperation, facilitating the exchange of critical alerts. By deploying time-sensitive processing techniques at the edge and utilizing DTs for comprehensive analysis, vehicles can take preemptive actions to avoid accidents. Road potholes contribute to traffic disruptions, the accordion effect, and vehicle damage, making their detection essential. This work explores advanced computer vision techniques implemented at the edge, specifically on vehicles. Roadside units (RSUs) offload DT data, and edge detection results are updated on the DT, providing a control center with accurate road condition information and maintaining a precise virtual replica of the environment. This distributed edge intelligence (DEI) enables rapid decision making with reduced latency while offering a comprehensive view of vehicle lifecycle management through DT data. The proposed algorithm, tested in real time, achieves mean average precision of 85% using YOLOv9t with minimal latency of 3 ms, ensuring effective pothole detection and seamless communication among nearby vehicles.**

***Index Terms*—Computer vision, digital twin (DT), edge computing, intelligent transportation system (ITS), Internet of Vehicles (IoV).**

I. Introduction

With the latest advancements in technology, the transportation sector has been significantly revolutionized, transitioning from human-driven to teleoperated and autonomous vehicles. However, the growing population and the inherent risks associated with high-speed travel necessitate innovative solutions for the transportation industry. According to the World Health Organization (WHO), 1.19 million people died in road traffic crashes in 2021, with 92% of these fatalities occurring in low- and middle-income countries, despite these countries having only around 60% of the world's vehicles [1]. A critical challenge, particularly evident in developing countries but increasingly recognized in developed nations as well, is the ongoing maintenance and repair of road infrastructure. Roads are often subjected to degradation over time due to heavy traffic loads, adverse weather conditions, and, in some cases, substandard construction practices. This degradation manifests in the formation of potholes, which pose significant risks to road safety and traffic flow. Potholes not only lead to serious accidents and traffic disruptions but also exacerbate the concertina effect, where the deceleration of one vehicle prompts a chain reaction of slowing vehicles, further increasing the risk of collisions.

Smart transportation systems and the Internet of Vehicles (IoV) have emerged as crucial solutions to address the challenges in transportation, making it safer, more reliable, and ensuring the smooth flow of traffic. IoV aims to create a framework where vehicles, road infrastructure, and communication devices are interconnected to enhance traffic efficiency, reduce environmental impacts, and, most importantly, ensure the safety of passengers and pedestrians. By deploying cutting-edge technologies, such as artificial intelligence (AI), machine learning (ML), vehicular ad-hoc networks (VANETs), digital twin (DT), and distributed edge intelligence (DEI), IoV enables real-time processing and decision making. This interconnected ecosystem facilitates seamless communication between Vehicle to Vehicle (V2V) and Vehicle to Infrastructure (V2I), leading to smarter and more responsive transportation systems, providing ubiquitous connectivity throughout the journey.

DEI, though still a relatively new concept, holds immense potential as a key enabler of the IoV and intelligent transportation systems (ITSs). The fundamental idea of edge intelligence is to bring computational power closer to the nodes, i.e., vehicles, rather than transferring all data to the cloud, they process and analyze data locally. This approach addresses critical issues of privacy and security while also solving latency problems, which are crucial in the vehicular ecosystem. By equipping end nodes with intelligence, critical decisions can be made without delay, reducing dependency on the cloud. Another key enabler of ITS is the DT. DT creates a virtual replica of real-world scenarios to perform real-time analysis, prognostic assessments, lifecycle management, and system performance optimization. The high fidelity in the modeling process and the integration of real-time data into

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the DT make it a cornerstone of the Industry 4.0 revolution across various sectors. DT utilizes Internet of Things (IoT), big data, and advanced communication systems to function effectively. The large volume of data needs to be continuously transmitted to the model for deep analysis. Continuous data transmission to the model is crucial for in-depth analysis. Here, roadside units (RSUs) play a vital role by offloading substantial data from vehicles, reducing the transmission load. Edge intelligence further optimizes this process by handling data at the edge node, transferring only essential outputs. This seamless integration of edge computing and DT enhances the precision and timeliness of analyses, ensuring proactive responses to potential issues and significantly improving safety and efficiency in transportation systems.

Using these key enabling technologies in combination with computer vision can achieve remarkable results. Computer vision, a field of AI that trains computers to interpret and understand the visual world, plays a critical role in analyzing and processing image data to make informed decisions. This technology uses deep learning (DL) models to identify and classify objects, detect anomalies, and extract valuable insights from visual inputs. One pressing issue that requires immediate attention is the detection of potholes on highways and roads. With the integration of cutting-edge technologies, such as DEI, DT, V2V, and V2I communication, vehicles can make informed decisions in real time. Cooperative perception and advanced computer vision techniques enable high-precision pothole detection and timely alerts to other drivers about potential dangers. This approach not only optimizes resource utilization but also enhances the overall reliability and resilience of the IoV ecosystem, making it a robust solution for the demands of modern transportation networks.

II. Literature Review

A critical aspect of ITS is the capacity to make instantaneous decisions, as even a brief delay can result in severe consequences. As a result, significant research is focused on edge computing and DEI to ensure rapid, secure, and reliable data processing. Edge computing underpins Industry 5.0 by facilitating the implementation of AI algorithms and models at the edge, thus enabling immediate data processing and reducing reliance on cloud-based computing [2]. Significant efforts have been made to enhance the safety and security of edge intelligence [3], [4], supporting the critical need for prompt and accurate decision making in ITS. In addition, advanced edge task offloading schemes have been proposed, enhancing processing capabilities and addressing security concerns [5], [6], [7]. Edge computing has paved the way for new technologies such as edge cloud computing, including vehicular cloud and multiaccess edge computing [8], [9]. The surge in research in this area is attributed to its tremendous processing power at the node, overcoming communication delays and security threats associated with transferring data to the cloud, making it ideal for ITS.

In both autonomous and manually driven vehicles, the integration of computer vision systems with various sensors provides a sophisticated solution for advanced driver-assistance systems (ADASs). This integration enhances decision making in critical driving scenarios, with advanced DL techniques improving ADAS accuracy and responsiveness [10]. With advancements in computational power, big data analytics, and the IoT, the accuracy of computer vision applications has significantly increased. Nevertheless, a key challenge in computer vision is the computational intensity of processing large volumes of data, necessitating the development of efficient algorithms for timely processing. Among the latest computer vision techniques, the you only look once (YOLO) framework has demonstrated state-of-the-art performance in object detection tasks, proving particularly suitable for real-time vehicular applications [11].

The presence of potholes is a significant concern for vehicle health, acting as a slow poison that deteriorates the condition of vehicles over time. The severity of potholes on vehicle repair can be estimated from statistics available from the American Automobile Association (AAA), which indicate that American drivers spend an average of $3 billion annually on pothole damage repairs [12]. This issue is not limited to America; it crosses borders. Statistics show that the pothole crisis costs England alone £1.49 billion annually in vehicle damage, as reported by the Centre for Economics and Business Research (CEBR) [13]. The issue persists in other parts of the world as well. In addition to financial losses, potholes contribute to accidents; in South Korea, 221 pothole-related accidents were reported from 2017 to 2019 [14]. The smooth flow of traffic is disrupted by the presence of potholes, causing a chain reaction of deceleration that amplifies congestion across the roadway. A study [15] highlighted that the presence of potholes over a 100-m segment can reduce vehicle speed by 55% and increase environmental emissions by 2.9% over the segment. Hence, this problem incurs not only financial losses and human casualties but also reduces vehicle lifespan and has adverse effects on the environment.

Various research studies have been conducted to detect potholes through different techniques, including vibration sensor-based approaches [16], thermal imaging [17], and autonomous aerial vehicles (AAVs) [18]. While each of these studies contributes valuable insights, they often require additional infrastructure and lack quick and accurate measures. With advancements in automotive technology, in-vehicle cameras have been utilized for pothole detection using vision-based techniques [19], [20], [21]. Various computer vision methods have been proposed, each striving to achieve high accuracy while maintaining lightweight models.

The study by [19] evaluated six YOLO versions and SSD-mobilenetv2 for pothole detection, proposing edge implementation within vehicles for minimal latency. Accuracy ranged from 0.47 to 0.95, with YOLOv5 showing promise but high latency. Tiny-YOLOv4, with a mean average precision (mAP) of 0.804 and a 96% confidence score, was deemed suitable for real-time detection. Park et al. [20] tested YOLOv4, YOLOv4-tiny, and YOLOv5 on the same dataset, achieving mAP scores of 0.777, 0.787, and 0.748, respectively. YOLOv4-tiny excelled in mAP but struggled with small potholes under adverse conditions, without real-world testing. The research by [22] used edge detection and YOLOv1,

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achieving mAP of 0.78 for single potholes, though it faced challenges in complex scenarios. Inference time and computational complexity were not discussed. Gajjar et al. [21] compared Faster R-CNN, SSD, and YOLOv3 for real-time detection, recommending Faster R-CNN for its reduced inference time despite its overall high latency. Sathya et al. [23] further examined YOLO models (v1, v2, and v3) with mAPs of 0.79–0.83, noting YOLOv3's promising results but lacking real-world data evaluation. Kumari et al. [24] assessed five YOLOv8 variants (YOLOv8n to YOLOv8x), with mAPs from 0.72 to 0.78. YOLOv8l and YOLOv8x performed best, but the study did not consider inference time or real-world testing.

While edge computing and computer vision effectively address the issue of timely action in real-world scenarios, the question arises: is a DT necessary in this framework? The primary purpose of DTs in Industry 4.0 and Industry 5.0 is to maintain a true replica at the virtual end, which is crucial for preemptive decision making, health life monitoring, and leveraging past experiences [25]. DTs have been widely incorporated in transportation for vehicle health monitoring, analyzing traffic trends, and monitoring complete life cycle performance [26], with ongoing work to enhance the security of DTs [27]. Researchers have also investigated using DTs for the detection of pavement cracks, showcasing its potential to accurately represent and monitor physical infrastructure [28].

In this study, we integrate the latest version of computer vision technology at the edge with DT frameworks, a novel approach that not only facilitates the rapid detection of potholes but also provides an accurate virtual representation of the physical entity. The primary contributions of this research are summarized as follows:

1) integrating DEI in the IoV ecosystem to enhance real-time pothole detection, thereby improving road safety, and facilitating vehicle maintenance;
2) implementing state-of-the-art computer vision techniques at the edge, i.e., within the vehicle, to achieve real-time, accurate detection and classification of potholes with minimal latency;
3) developing a comprehensive DT of the vehicle by collecting data from sensors through RSUs and update the road condition virtually from the processed data;
4) utilizing cooperative perception to share pothole presence using V2V and V2I communication within VANET, sending alert messages to the augmented reality head-up display (AR-HUD) of nearby vehicles and notifying control center of the pothole location.

III. Pothole Detection Framework

A. System Model

Fig. 1 illustrates the overall framework of our approach for pothole detection in ITS employing DEI. In this scenario, four cars are shown moving on a road with potholes. Each car in the network is equipped with an onboard unit (OBU), dedicated short-range communication (DSRC), and a dashcam. These vehicles act as edge devices, forming a distributed edge network with enhanced capabilities for performing intelligent tasks locally. Utilizing cooperative perception, they can share information, enabling timely actions by incoming vehicles.

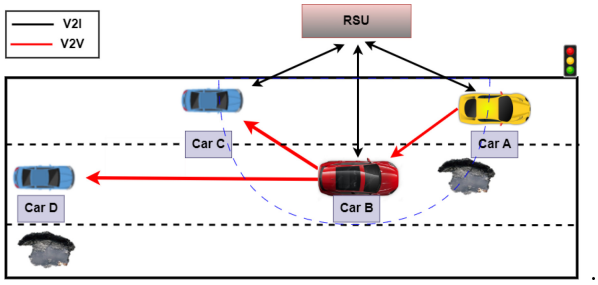


Fig. 1. Framework for pothole detection in ITS employing edge intelligence.

TABLE I
Distribution of Sensors and Data Columns for Vehicle Monitoring

Monitored Component	No. of Data Columns
Steering angle	1
Brake pedal	1
Speed	1
Longitudinal acceleration	1
Lateral acceleration	1
Position	3
Rotation	9
Front left wheel	16
Front right wheel	16
Rear left wheel	16
Rear right wheel	16

As depicted in Fig. 1, Car A processes its dashcam video data locally and detects a pothole. First, it alerts its driver by displaying a warning message on the vehicle's AR-HUD. It then sends this information to Car B and the RSU. Car B, having received the pothole alert from Car A via V2V communication using DSRC, can now take necessary precautions and relay this information to other vehicles through the vehicular network (V-Net). Communication between vehicles and the RSU is bidirectional, as shown in the figure. All vehicles within the RSU's vicinity, illustrated by the blue dotted line, offload their real-time sensor data to the RSU, reducing the computational load on edge devices (i.e., vehicles), and receive pothole presence alerts. Cars not in the vicinity of the RSU will receive alerts on their AR-HUD from neighboring vehicles, as illustrated with Vehicle D in the figure. This cooperative perception and edge intelligence ensure that accurate and timely information is passed to other traffic, maintaining the flow of traffic.

1) Mathematical Model of the Vehicle Digital Twin: The DT model comprises a physical vehicle equipped with 81 sensors located at various assemblies of the vehicle. Details of these sensors are provided in Table I. The vehicle periodically collects data from these sensors and offloads this substantial data to the nearest RSUs within its vicinity. The data is then transferred to a remote server, where simulations and in-depth analyses are performed, including detailed lifecycle analysis, prognostic analysis, and comprehensive evaluations of traffic management.

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Our simulations utilize the principles of a nonlinear gray box vehicle dynamics model [29], which served as the foundation for constructing a Simulink model of the vehicle. This model was then translated into a virtual reality environment using virtual reality markup language (VRML) for integration with the DT. The dynamic behavior of the vehicle, including its longitudinal speed $u_x(t)$, lateral speed $u_y(t)$, and yaw rate $\omega(t)$, around its center of gravity, is described by the following set of ordinary differential equations:

$$\begin{aligned}\frac{d}{dt}u_x(t) = {} & u_y(t)\cdot\omega(t) + \frac{1}{m}\Big[\big(\Phi_{x,T_{\mathrm{FL}}}(t) + \Phi_{x,T_{\mathrm{FR}}}(t)\cos\theta(t)\big) \\ & - \big(\Phi_{y,T_{\mathrm{FL}}}(t) + \Phi_{y,T_{\mathrm{FR}}}(t)\sin\theta(t)\big) + \Phi_{x,T_{\mathrm{RL}}}(t) \\ & + \Phi_{x,T_{\mathrm{RR}}}(t)\Big] - C_A\cdot u_x(t)^2 \qquad (1)\\ \frac{d}{dt}u_y(t) = {} & -u_x(t)\cdot\omega(t) + \frac{1}{m}\Big[(\Phi_{x,T_{\mathrm{FL}}}(t) + \Phi_{x,T_{\mathrm{FR}}}(t)\sin\theta(t)) \\ & + \big(\Phi_{y,T_{\mathrm{FL}}}(t) + \Phi_{y,T_{\mathrm{FR}}}(t)\cos\theta(t)\big) \\ & + \Phi_{y,T_{\mathrm{RL}}}(t) + \Phi_{y,T_{\mathrm{RR}}}(t)\Big] \qquad (2)\\ \frac{d}{dt}\omega(t) = {} & \frac{1}{J}\Big[a\cdot\Big(\big(\Phi_{x,T_{\mathrm{FL}}}(t) + \Phi_{x,T_{\mathrm{FR}}}(t)\sin\theta(t)\big) \\ & + \big(\Phi_{y,T_{\mathrm{FL}}}(t) + \Phi_{y,T_{\mathrm{FR}}}(t)\cos\theta(t)\big)\Big) \\ & - b\cdot\big(\Phi_{y,T_{\mathrm{RL}}}(t) + \Phi_{y,T_{\mathrm{RR}}}(t)\big)\Big]. \qquad (3)\end{aligned}$$

In these equations, the force components Φ_x and Φ_y are associated with the longitudinal and lateral directions, respectively. The subscripts T_{FL}, T_{FR}, T_{RL}, and T_{RR} denote the front left, front right, rear left, and rear right tires, respectively. The steering angle is indicated by $\theta(t)$, and J signifies the moment of inertia. The parameters a and b represent the distances from the center of gravity to the front and rear axles, respectively.

The linear approximations employed for the tire forces are as follows, for $i \in \{T_{\mathrm{FL}}, T_{\mathrm{FR}}, T_{\mathrm{RL}}, T_{\mathrm{RR}}\}$:

$$\Phi_{x,i}(t) = D_x\cdot s_i(t) \qquad (4)$$

$$\Phi_{y,i}(t) = D_y\cdot\beta_i(t). \qquad (5)$$

In (4) and (5), D_x and D_y represent the longitudinal and lateral tire stiffness, respectively.

The front and rear tire slip angles are approximated by the following equations. Here, $k \in \{L, R\}$, where L represents the left tire and R represents the right tire

$$\beta_{T_{\mathrm{F}},k}(t) = \theta(t) - \left(u_y(t) + a\cdot\frac{\omega(t)}{u_x(t)}\right) \qquad (6)$$

$$\beta_{T_R,k}(t) = -u_y(t) - b\cdot\frac{\omega(t)}{u_x(t)}. \qquad (7)$$

The equations presented form the foundation for the state-space modeling of the system. In this framework, the system states $x_1(t)$, $x_2(t)$, and $x_3(t)$ are represented by the longitudinal speed $u_x(t)$, lateral speed $u_y(t)$, and yaw rate $\omega(t)$, respectively. The inputs to the state-space model are $s_{T_{\mathrm{FL}}}(t)$, $s_{T_{\mathrm{FR}}}(t)$, $s_{T_{\mathrm{RL}}}(t)$, $s_{T_{\mathrm{RR}}}(t)$, and $\theta(t)$, where $s_{T_{\mathrm{FL}}}(t)$, $s_{T_{\mathrm{FR}}}(t)$, $s_{T_{\mathrm{RL}}}(t)$, and $s_{T_{\mathrm{RR}}}(t)$ represent the slip values of the front left, front right, rear left, and rear right tires, respectively, and $\theta(t)$ is the steering angle. The outputs of the system are the longitudinal speed $u_x(t)$, lateral acceleration $a_y(t)$, and yaw rate $\omega(t)$.

2) Vehicle Dynamics Dataset: The proposed pothole detection framework integrates 81 distinct sensor readings from the vehicle to construct a DT remotely. These readings are captured every 0.04 s over a 110-s period from a Skoda Octavia. The implementation utilizes advanced tools and libraries, including the Computer Vision Toolbox, Simulink 3-D Animation, and MATLAB V-Realm Builder 2.0. The sensors are strategically positioned to measure various parameters, including antilock braking system (ABS) status, powertrain control, chassis dynamics, damper displacements, temperature variations, steering angles, and tire rotations at both the front and rear. These measurements, detailed in Table I, are essential for developing an accurate digital replica of the vehicle, optimizing performance, and facilitating comprehensive analysis. At the DT interface, a sophisticated user experience is provided through 2-D graphical representations and VR holograms, accurately simulating real-world scenarios and preserving a comprehensive lifecycle history.

B. Computer Vision Techniques

For our task of timely detection of potholes using dashcam footage, we employed the advanced computer vision technique YOLO. YOLO is a real-time object detection method that uses a single neural network to analyze the entire image, delivering fast and accurate predictions by considering the full image context. YOLO has been publicly available since its inception 2015, with each update enhancing its performance. YOLOv9 introduced features like programmable gradient information (PGI) and the generalized efficient layer aggregation network (GELAN), offering seven variants to accommodate different processing and latency requirements [30]. The latest YOLOv10, released in May 2024, further improves real-time detection by optimizing components and eliminating nonmaximum suppression (NMS), achieving superior performance on the COCO dataset [31]. It provides six versions, from Nano (n) to Extra Large (X), balancing accuracy and response time for various applications.

For our work on pothole detection in resource-constrained environments, i.e., at the edge, we require an algorithm that balances low computational cost with minimal latency and high accuracy. Consequently, we selected three of the latest YOLO variants: YOLOv9t, YOLOv9s, and YOLOv10n. These variants were chosen due to their efficient performance in real-time detection, their ability to handle varying levels of computational resources, and their effectiveness in accurately identifying potholes while maintaining low latency.

1) Training Dataset: For pothole detection, we utilize the publicly available dataset [32]. This dataset includes images of potholes, varying from single to multiple instances per frame, captured by dashcams under diverse lighting conditions and times of day, to ensure robust model training. To further enhance the dataset, augmentation techniques, such as random cropping, rotation, shearing, and adjustments in brightness and exposure, were applied to the training set. All images were preprocessed to a consistent size of 640 × 640 pixels. Each image was annotated using the YOLO format, which specifies bounding box coordinates in normalized form relative to the

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image dimensions. The annotation file includes the object class (i.e., pothole) along with the normalized coordinates for the center of the bounding box and its dimensions.

2) Simulation Setup: The simulations were conducted using Ultralytics, implemented in Python 3.10.12, on a machine with an Intel Core i5-1035G1 CPU @ 1.00 GHz (1.19 GHz) and 12.0 GB of RAM. We employed transfer learning to train the model. Since the COCO dataset, which contains 80 categories, does not include potholes, we fine-tuned YOLO by freezing its pretrained layers and retraining the final layer specifically for pothole detection. This approach utilizes the features learned from the COCO dataset while adapting the model to the specific task of pothole detection. All three models were trained on the same dataset with consistent training parameters, including 100 epochs, a batch size of 16, an initial learning rate of 0.0001, a final learning rate of 0.01, dropout regularization set to 0.25, and an auto optimizer. This approach ensured a fair comparison and facilitated the selection of the most effective model for pothole detection.

IV. Methodology

To implement the proposed pothole detection system at the edge and update the DT remotely, we adopted a systematic approach integrating advanced computer vision techniques and real-time data processing. We evaluated three variants of YOLO—YOLOv9t, YOLOv9s, and YOLOv10n—to identify the model that offers the optimal balance of minimal latency and high accuracy at the edge device. The model that outperformed the others was YOLOv9t, as discussed in Section V. Pothole detection performed at the edge is not only communicated to neighboring vehicles but also sent to the RSU. The RSU offloads the DT voluminous sensor readings along with the processed information, ensuring a true replica of the vehicle's health and updating the road condition. If the number of vehicles reporting a detected pothole exceeds a threshold β (set to 50 in our work), the DT updates its virtual roadmap to incorporate the identified pothole. This approach helps mitigate false positives and false negatives by leveraging aggregated data. The overall framework of the pothole detection and notification system is outlined in Algorithm 1.

Algorithm 1 Pothole Detection and Notification System

Require: Video feed from vehicle camera, VehicleDynamics.csv (81 features), Edge Processing Unit (EPU), Roadside Unit (RSU), YOLO model
Ensure: Timely notification of potholes to vehicles and authorities
Initialization:
Load YOLO model for pothole detection
Load DT model for real-time updates
Set $T_{conf} = \alpha$ (confidence score threshold)
Set $T_{count} = \beta$ (vehicle count threshold)
Data Collection and Edge Processing:
while vehicle is in motion **do**
    Capture video frames
    Resize images to 640 × 640
    Collect VehicleDynamics.csv and GPS location
    Pothole Detection:
    Apply YOLO model
    if confidence score > T_{conf} **then**
        Record pothole data (location (GPS), count)
        Transmit video frames, VehicleDynamics.csv, and pothole data to RSU
        Broadcast pothole info to nearby vehicles
    else
        Transmit VehicleDynamics.csv
    end if
end while
RSU Processing and DT Update:
for each received data **do**
    Update DT model with received vehicle dynamics data
    Increment pothole counter for detected locations
    if pothole counter > T_{count} **then**
        Mark central GPS point on DT
        Update virtual map with pothole location
        Notify authorities
    end if
    Notify approaching vehicles via RSU
end for
Vehicle Notification:
if pothole information received from RSU or nearby vehicles **then**
    Activate buzzer
    Display AR-HUD message: *"Pothole ahead!"*
    Broadcast pothole info to vehicles within range
end if

V. Results and Discussion

A. Pothole Detection Algorithm at the Edge

The edge computing paradigm is employed for real-time pothole detection, leveraging three variants of the YOLO algorithm: YOLOv9t, YOLOv9s, and YOLOv10n. Each model's performance is evaluated using the same parameters mentioned in Section III-B2, using metrics, such as mAP and inference time. The mAP is defined as the average precision across all detection thresholds, which can be formulated as follows:

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n} \text{AP}_i \tag{8}$$

where n is the number of classes and AP_i is the average precision for class i.

The results, shown in Fig. 2, highlight the strengths and weaknesses of each model in an edge environment. Each model's performance under the same conditions is detailed below.

1) YOLOv9t: The mAP@0.5 for YOLOv9t is 0.82. The inference time is 3.1 ms, making it suitable for scenarios that require high precision with low latency. This balance of accuracy and speed demonstrates the model's effectiveness in real-time applications.

2) YOLOv9s: The mAP@0.5 for YOLOv9s is 0.77, with an inference time of 6.7 ms. While this model provides good accuracy, its higher inference time makes it less suitable for real time application like pothole detection.

3) YOLOv10n: The mAP@0.5 for YOLOv10n is 0.69, and the inference time is 3 ms. This model offers a reasonable tradeoff between accuracy and speed, but it does not outperform YOLOv9t in terms of precision.

Among all these models, YOLOv9t outperforms the other models in our case due to its reduced latency and highest

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TABLE II
COMPARISON OF YOLO VARIANTS FOR POTHOLE DETECTION

Technique	Layers	Inf. Time (ms)	mAP @0.5	GFLOPs	Params
YOLOv9t	486	3.1	0.82	7.6	1,971k
YOLOv9s	486	6.7	0.77	26.7	7,167k
YOLOv10n	285	3.0	0.69	8.2	2,694k

accuracy, making it the best choice for edge-based real-time pothole detection. As shown in Table II, YOLOv9t achieved an inference time of 3.1 ms and a mAP@0.5 of 0.82 with 7.6 GFLOPs and 1971k parameters, outperforming YOLOv9s and YOLOv10n in both speed and accuracy. This makes YOLOv9t particularly suitable for applications requiring real-time processing and prompt hazard alerts to enhance road safety.

The overall computational complexity of the YOLO models, including YOLOv9 and YOLOv10, can be expressed as

$$O\left(n^2 \times C \times (5B + k)\right) \tag{9}$$

where n is the input size, C is the number of grid cells, B is the number of bounding boxes per cell, and k is the number of classes. This complexity formula captures the full detection process, including the grid cell division and bounding box prediction.

The result of pothole detection on the validation dataset using YOLOv9t is shown in Fig. 3. Further evaluation was conducted in real-world scenarios in Pakistan over a period of one month, where representative samples are depicted in Fig. 4. The mAP on the test set in these real-world conditions was found to be 0.85.

B. Visual Analysis at the DT

The proposed system detects the presence of potholes at the edge using the YOLO algorithm. Once a pothole is detected, this information, along with other vehicle dynamics data, is sent to the nearest RSU. The data sent includes the number of potholes and their GPS locations. This processed information, collected from the vehicles, is then fed into the DT model. Given that the accuracy of GPS is approximately 5 m, the DT model employs a threshold-based mechanism for reliability. If the number of potholes reported by multiple vehicles within a GPS radius of 5 m exceeds the threshold β (set to 50 in our simulations), and the confidence score for each detection is greater than α (set to 0.75 in our work), a central point is marked on the DT visualization to represent the detected pothole, as depicted in Fig. 5. The threshold value β, was determined to optimize detection accuracy and practicality, guided by preliminary testing and aligned with crowd-sourcing practices from prior studies [33]. This collective consensus policy ensures that the reported pothole existence by multiple vehicles is highly reliable and minimizes false positives.

C. In-Vehicle Visual Alerts

For real-time pothole detection within the vehicle, timing is crucial. To provide timely alerts about potholes, each

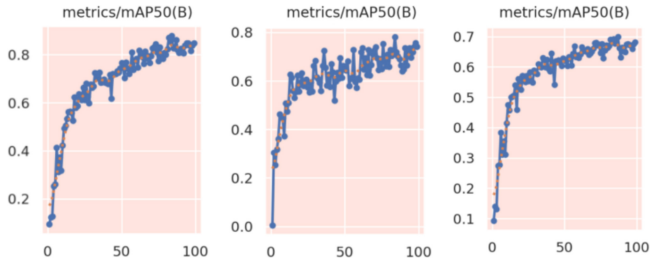


Fig. 2. Comparison of mAP across 100 epochs for YOLOv9t, YOLOv9s, and YOLOv10n (left to right).



Fig. 3. Evaluation of the YOLOv9t algorithm for pothole detection on the validation dataset.



Fig. 4. Assessment of the YOLOv9t algorithm for pothole detection on a real-world dataset collected during both sunny and cloudy days.

vehicle is equipped with a dashcam that processes video footage and delivers results promptly. Frames were extracted from the video at a fixed sampling rate of 30 frames per second (FPS), a widely adopted standard in real-time video processing that ensures smooth motion visualization, compatibility with camera hardware, and efficient processing without overloading computational resources. To enhance the robustness and efficiency of the framework, we propose a collaborative learning approach. Through DSRC, the presence of a pothole detected by one vehicle is communicated to following vehicles and nearby RSUs. This V2I, V2V, and edge communication ensures that vehicles receive timely notifications, mitigating potential damage—especially in scenarios where the dashcam might fail to accurately detect potholes due to lighting conditions. V2I communication is particularly vital as it disseminates information collected from the DT system. Each vehicle, equipped with an AR-HUD controller, will display an alert message on the driver's screen, making them aware of upcoming road conditions.

VI. PERFORMANCE VALIDATION

To evaluate the effectiveness and innovation of our pothole detection approach, we systematically compared it with existing state-of-the-art methods. This evaluation encompassed a range of computer vision techniques employed for pothole detection, focusing on accuracy, latency, and their applicability

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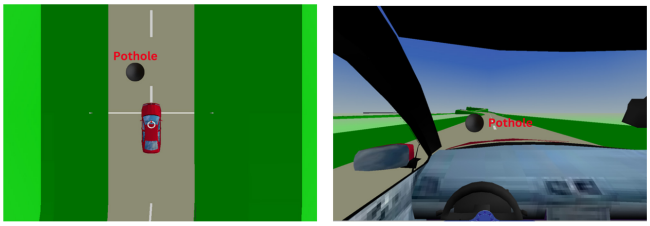


Fig. 5. Visualization of pothole detection and reporting in the DT System. The left figure shows the top view of the car with detected potholes within the DT model, while the right figure illustrates the driver's perspective at the DT with highlighted pothole alerts.

TABLE III
COMPARISON WITH STATE-OF-THE-ART POTHOLE DETECTION TECHNIQUES

Ref:	Technique:	mAP@0.5:	Inf. Time (ms):	Limitation:
[19]	YOLOv1, YOLOv2, YOLOv3, YOLOv4, Tiny-YOLOv4, YOLOv5, SSD-mobilenetv2	0.795, 0.812, 0.836, 0.855, 0.804, 0.950, 0.474	340, 33.7, 70.57, 52.51, 4.86, 7	Inference time is high for real time detection on the edge.
[20]	YOLOv4, Tiny-YOLOv4, YOLOv5s	0.777, 0.787, 0.748	-	Computationally intensive, with 4000-6000 training epochs and 720 × 720 input images. The study lacked inference time details and evaluation under diverse weather conditions.
[22]	YOLOv1	0.780	-	Limited to detecting a single pothole, the study utilized low-resolution input images (486 × 353), but did not address computational complexity.
[21]	Faster RCNN, SSD, YOLOv3	0.415, 0.185, 0.347	7020, 4815, 836	A diverse dataset was used for testing, but inference time was high due to the 1280 × 720 resolution images, with training epochs ranging from 1200 to 6000.
[23]	YOLOv1, YOLOv2, YOLOv3	0.797, 0.822, 0.833	-	Computational complexity is not discussed.
[24]	YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, YOLOv8x	0.782, 0.727, 0.787, 0.787, 0.785	-	Inference time is not mentioned. Not evaluated on a real real-world environments.
Our Work	YOLOv9t, YOLOv9s, YOLOv10n	0.824, 0.773, 0.69	3.1, 6.7, 3	Computationally lightweight with accuracy ensured on testing data collected from Dashcam.

to real-world datasets. The comparative results are summarized in Table III.

Our implementation of YOLOv9t was assessed against approaches such as those presented by [19], which examined multiple YOLO versions and SSD-mobilenetv2. Notably, YOLOv9t demonstrated enhanced performance, particularly in inference time. While Park et al. [20] reported a high mAP of 0.787 for YOLOv4-tiny, this approach encountered difficulties with detecting small potholes under adverse conditions and lacked real-world testing. Our method effectively mitigates these challenges. Additionally, the approach described in [22], which utilized edge detection with YOLOv1, achieved mAP of 0.78 but was limited by its handling of complex scenarios and did not address inference time or computational complexity. Our method improves upon these limitations by ensuring real-time detection with optimized latency. Moreover, Gajjar et al. [21] recommended Faster R-CNN for its reduced inference time, although it exhibited a higher overall inference time compared to our approach. Further comparative analysis includes [23], which evaluated three YOLO variants on a custom dataset. Although YOLOv3 demonstrated promising results, the study did not address computational complexity or real-world application. Similarly, Kumari et al. [24] assessed YOLOv8 variants without considering inference time or real-world testing. In contrast, our YOLOv9t implementation has been rigorously validated on real-world datasets, as depicted in Fig. 4, offering robust and reliable performance.

This comparative analysis underscores the superiority of our approach over existing methods, particularly in real-world scenarios, where our model achieves the highest accuracy with minimal latency, addressing the tradeoffs seen in previous models that offered competitive accuracy but suffered from excessive inference time.

VII. CONCLUSION

The primary aim of this study is to address road safety and enhance cooperative perception for pothole detection in ITS. This article has demonstrated that integrating computer vision and DT technologies is essential for improving ITS' safety and efficiency. High-speed vehicles have required timely alerts to prevent collisions, with ADAS and VANETs facilitating critical IoVs' communication. By implementing advanced computer vision techniques at the edge and utilizing DTs for real-time analysis, vehicles can take preemptive actions to avoid accidents through effective pothole detection. The proposed algorithm, tested in real time, has achieved mAP of 85% with minimal latency of 3 ms, ensuring seamless communication and rapid decision making. This research presents a considerable foundation for future work to expand the framework for detecting other road irregularities, including speed bumps, debris, and slippery surfaces, turning the framework even more valuable for practical application. Furthermore, we aim to integrate federated learning to enhance model performance during interactions between edge devices and DTs in IoV, ensuring data privacy and strengthening the system's resilience against potential cyber threats.

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